MIDDLESEX SCHOOL | MATHEMATICS

Math 12 Course at a Glance

Intermediate Algebra

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M12-REV-001	I can evaluate numerical expressions using the order of operations.
M12-REV-002	I can perform operations with signed integers and fractions.
M12-REV-003	I can simplify an algebraic expression by combining like terms.
M12-REV-004	I can use the distributive property to rewrite an expression.
M12-REV-005	I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED	
M12-EQN-001	I can solve a linear equation and present equivalent steps clearly.
M12-EQN-002	I can solve a linear equation containing fractions.
M12-EQN-003	I can rearrange a formula to isolate a specified variable.
M12-EQN-004	I can translate a verbal relationship into a linear equation.
M12-EQN-005	I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED	
M12-INE-001	I can solve and graph a one-variable linear inequality.
M12-INE-002	I can solve a compound inequality involving AND.
M12-INE-003	I can solve a compound inequality involving OR.
M12-INE-004	I can express an inequality solution using interval notation.
M12-INE-005	I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED	
M12-LIN-001	I can determine and interpret the slope of a line.
M12-LIN-002	I can graph a line from an equation in a common form.
M12-LIN-003	I can write an equation of a line from given information.
M12-LIN-004	I can determine whether two lines are parallel, perpendicular, or neither.
M12-LIN-005	I can evaluate a function using function notation.
M12-LIN-006	I can determine whether a relation is a function using the vertical line test or input-output data.
M12-LIN-008	I can determine domain and range from a graph or table.
M12-LIN-009	I can interpret domain and range in context.

Math 12 Course at a Glance

Intermediate Algebra

Systems of Linear Equations

REQUIRED	
M12-SYS-001	I can interpret the solution of a linear system as an intersection point.
M12-SYS-002	I can solve a two-variable linear system by graphing.
M12-SYS-003	I can solve a two-variable linear system by substitution.
M12-SYS-004	I can solve a two-variable linear system by elimination.
M12-SYS-005	I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED	
M12-POL-001	I can simplify expressions using integer exponent rules.
M12-POL-002	I can add and subtract polynomials.
M12-POL-003	I can multiply polynomials, including binomials.
M12-POL-004	I can divide a polynomial by a monomial.
M12-POL-005	I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION	
M12-EXT-001	I can solve and graph an absolute-value inequality.
M12-EXT-002	I can graph a system of linear inequalities and identify its solution region.
M12-EXT-003	I can factor a quadratic trinomial.
M12-EXT-004	I can identify a relationship as discrete or continuous.

MIDDLESEX SCHOOL | MATHEMATICS

Math 21 Course at a Glance

Algebra and Its Functions

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M21-REV-001	I can determine and interpret the slope of a line.
M21-REV-002	I can graph a line from an equation and write an equation from given information.
M21-REV-003	I can apply a linear function to a context.
M21-REV-004	I can identify the solution of a linear system as an intersection point.
M21-REV-005	I can solve a simple two-variable linear system using an appropriate method.
M21-REV-006	I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED	
M21-FUN-001	I can evaluate a function from a formula, graph, or table.
M21-FUN-002	I can determine whether a relation is a function.
M21-FUN-003	I can determine domain and range from a graph or table.
M21-FUN-004	I can solve for an input that produces a specified output.
M21-FUN-005	I can interpret inputs, outputs, domain, and range in context.
M21-FUN-006	I can calculate and interpret average rate of change.
M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED	
M21-POL-001	I can add and subtract polynomials.
M21-POL-002	I can multiply polynomials.
M21-POL-003	I can factor out a greatest common factor.
M21-POL-004	I can factor a quadratic trinomial.
M21-POL-005	I can factor a difference of squares and other required special cases.
M21-POL-006	I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED	
M21-RAT-001	I can state excluded values for a rational expression.
M21-RAT-002	I can simplify a rational expression while preserving its restrictions.
M21-RAT-003	I can multiply and divide rational expressions.
M21-RAT-004	I can add and subtract rational expressions using a common denominator.
M21-RAT-005	I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED	
M21-EXP-001	I can simplify expressions containing negative exponents.
M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
M21-EXP-003	I can simplify radical expressions.
M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Math 21 Course at a Glance

Algebra and Its Functions

Quadratic Functions and Equations

REQUIRED	
M21-QUAD-001	I can identify key features of a quadratic function from its graph or equation.
M21-QUAD-002	I can connect standard, factored, and vertex forms to features of a parabola.
M21-QUAD-003	I can solve a quadratic equation by factoring.
M21-QUAD-004	I can solve a quadratic equation using the square-root property.
M21-QUAD-005	I can solve a quadratic equation by completing the square.
M21-QUAD-006	I can solve a quadratic equation using the quadratic formula.
M21-QUAD-007	I can choose an efficient method for solving a quadratic equation.
M21-QUAD-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
M21-QUAD-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION	
M21-EXT-001	I can perform basic arithmetic with complex numbers.
M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.

Math 22 Course at a Glance

Geometry

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M22-REV-001	I can solve a linear equation arising from a geometric relationship.
M22-REV-002	I can solve a proportion and interpret the resulting scale factor.
M22-REV-003	I can simplify an exact square root and approximate it when needed.
M22-REV-004	I can calculate distance, midpoint, and slope on the coordinate plane.
M22-REV-005	I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED	
M22-BAS-001	I can name and represent points, lines, planes, segments, and rays using correct notation.
M22-BAS-002	I can identify collinear, coplanar, intersecting, and concurrent figures.
M22-BAS-003	I can calculate segment lengths using betweenness and the segment-addition postulate.
M22-BAS-004	I can determine a midpoint or use a segment bisector relationship.
M22-BAS-005	I can classify and measure angles.
M22-BAS-006	I can calculate angle measures using angle addition and angle bisectors.
M22-BAS-007	I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
M22-BAS-008	I can distinguish a definition, postulate, theorem, converse, and counterexample.
M22-BAS-009	I can determine angle relationships formed by parallel lines and a transversal.
M22-BAS-010	I can use angle relationships to determine whether lines are parallel.
M22-BAS-011	I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
M22-BAS-012	I can construct a logical geometric argument from stated facts and accepted results.

Math 22 Course at a Glance

Geometry

Triangle Congruence and Proof

REQUIRED	
M22-CON-001	I can classify triangles by sides and angles.
M22-CON-002	I can use triangle angle-sum and exterior-angle relationships.
M22-CON-003	I can write a congruence statement with corresponding vertices in the correct order.
M22-CON-004	I can determine whether triangles are congruent by SAS.
M22-CON-005	I can determine whether triangles are congruent by SSS.
M22-CON-006	I can determine whether right triangles are congruent by HL.
M22-CON-007	I can determine whether triangles are congruent by ASA or AAS.
M22-CON-008	I can apply isosceles and equilateral triangle theorems and their converses.
M22-CON-009	I can use corresponding parts of congruent triangles after proving congruence.
M22-CON-010	I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED	
M22-QUAD-001	I can calculate interior or exterior angle measures of a polygon.
M22-QUAD-002	I can calculate perimeter and area of common polygons.
M22-QUAD-003	I can apply properties of parallelograms to determine unknown measures.
M22-QUAD-004	I can determine whether a quadrilateral is a parallelogram.
M22-QUAD-005	I can apply properties of rectangles, rhombi, and squares.
M22-QUAD-006	I can classify a quadrilateral from its stated or marked properties.
M22-QUAD-007	I can apply trapezoid and trapezoid-midsegment relationships.
M22-QUAD-008	I can apply properties of kites.
M22-QUAD-009	I can calculate the perimeter or area of a composite polygon.
M22-QUAD-010	I can justify a quadrilateral conclusion using definitions and theorems.

Similarity and Proportional Reasoning

REQUIRED	
M22-SIM-001	I can determine image lengths and coordinates under a dilation.
M22-SIM-002	I can identify the scale factor of a dilation or similarity relationship.
M22-SIM-003	I can write a similarity statement with corresponding vertices in the correct order.
M22-SIM-004	I can use proportional corresponding sides to solve similar-polygon problems.
M22-SIM-005	I can relate similarity scale factor to perimeter and area scale factors.
M22-SIM-006	I can prove triangles similar by AA, SSS, or SAS.
M22-SIM-007	I can apply triangle proportionality theorems and their converses.
M22-SIM-008	I can solve and justify an indirect-measurement or other similarity application.

Math 22 Course at a Glance

Geometry

Right Triangles

REQUIRED

M22-RGT-001	I can apply the Pythagorean theorem to determine a missing side.
M22-RGT-002	I can use the converse of the Pythagorean theorem to classify a triangle.
M22-RGT-003	I can apply triangle-inequality relationships.
M22-RGT-004	I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
M22-RGT-005	I can apply right-triangle similarity relationships.
M22-RGT-006	I can use geometric-mean relationships created by an altitude to the hypotenuse.
M22-RGT-007	I can select sine, cosine, or tangent for a right-triangle problem.
M22-RGT-008	I can determine a missing side or angle of a right triangle.
M22-RGT-009	I can solve a contextual right-triangle problem.
M22-RGT-010	I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

M22-CIR-001	I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
M22-CIR-002	I can calculate circumference and arc length.
M22-CIR-003	I can calculate the area of a circle, sector, or segment.
M22-CIR-004	I can determine angle and arc measures involving central and inscribed angles.
M22-CIR-005	I can apply relationships among chords, arcs, and distances from the center.
M22-CIR-006	I can apply tangent-radius and tangent-segment relationships.
M22-CIR-007	I can determine angle measures formed by chords, secants, or tangents.
M22-CIR-008	I can determine segment lengths formed by intersecting chords.
M22-CIR-009	I can determine segment lengths formed by secants or tangents.
M22-CIR-010	I can solve a multi-step circle problem and justify each relationship used.

MIDDLESEX SCHOOL | MATHEMATICS

Math 31 Course at a Glance

Advanced Algebra

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M31-REV-001	I can calculate a percentage, percent increase, or percent decrease.
M31-REV-002	I can simplify an expression using exponent properties.
M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.
M31-REV-004	I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED	
M31-LIN-001	I can interpret function notation as a statement about input and output.
M31-LIN-002	I can determine whether a graph or table represents a function.
M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula.
M31-LIN-004	I can identify independent and dependent variables in a relationship.
M31-LIN-005	I can calculate average rate of change from two points.
M31-LIN-006	I can interpret interval notation and identify increasing or decreasing intervals.
M31-LIN-007	I can determine total change from an average rate of change over an interval.
M31-LIN-008	I can determine and interpret the units of an average rate of change.
M31-LIN-009	I can identify a linear function from a table, graph, or formula.
M31-LIN-010	I can interpret the slope and intercept of a linear model.
M31-LIN-011	I can determine the intersection of two linear relationships graphically or algebraically.
M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Math 31 Course at a Glance

Advanced Algebra

Function Behavior and Structure

REQUIRED

M31-FUN-001	I can determine whether a relation is a function.
M31-FUN-002	I can evaluate a function from a formula, graph, or table.
M31-FUN-003	I can determine domain and range from a formula, graph, or table.
M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots.
M31-FUN-005	I can graph a piecewise function from its formula.
M31-FUN-006	I can write a piecewise formula that represents a graph.
M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections.
M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes.
M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables.
M31-FUN-010	I can write a formula for a composition of two functions.
M31-FUN-011	I can find an inverse function algebraically.
M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table.
M31-FUN-013	I can determine and interpret the units of an inverse function.
M31-FUN-014	I can explain how the domain and range of a function relate to those of its inverse.
M31-FUN-015	I can determine the range of a function by analyzing the domain of its inverse.
M31-FUN-016	I can verify algebraically that two functions are inverses.
M31-FUN-017	I can interpret the meaning and units of a composition.
M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change.
M31-FUN-019	I can determine whether a function is invertible using the horizontal line test.

Function Behavior and Structure

M31-FUN-020	I can graph an inverse function as a reflection across the line `y = x`.

Quadratic Functions and Equations

REQUIRED

M31-QUAD-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.
M31-QUAD-002	I can connect standard, factored, and vertex forms to features of a quadratic graph.
M31-QUAD-003	I can rewrite a quadratic function in a form that reveals a requested feature.
M31-QUAD-004	I can solve a quadratic equation by factoring or using the quadratic formula.
M31-QUAD-005	I can graph a quadratic function using its algebraic structure and key features.
M31-QUAD-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information.
M31-QUAD-007	I can create and interpret a quadratic trajectory model.

Math 31 Course at a Glance

Advanced Algebra

Exponential Functions and Models

REQUIRED

M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change.
M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model.
M31-EXP-003	I can write an exponential function from a table, graph, or context.
M31-EXP-004	I can evaluate and interpret an exponential model.
M31-EXP-005	I can calculate growth under periodic compounding.
M31-EXP-006	I can calculate growth under continuous compounding.
M31-EXP-007	I can determine an exponential formula from two points or other sufficient data.
M31-EXP-008	I can determine domain, range, intercepts, and asymptotes of an exponential function.
M31-EXP-009	I can compare the growth rates of two exponential functions.
M31-EXP-010	I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

M31-LOG-001	I can convert between logarithmic and exponential forms.
M31-LOG-002	I can evaluate a logarithm using its exponential meaning.
M31-LOG-003	I can expand or condense an expression using product, quotient, and power properties.
M31-LOG-004	I can solve an exponential equation using logarithms.
M31-LOG-005	I can solve a logarithmic equation using exponential form and check domain restrictions.
M31-LOG-006	I can calculate and interpret doubling time or half-life.
M31-LOG-007	I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
M31-LOG-008	I can explain how exponential and logarithmic functions undo one another.
M31-LOG-009	I can solve a multi-step exponential or logarithmic equation.
M31-LOG-010	I can determine an intersection of exponential functions graphically or algebraically.
M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model.
M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

M31-EXT-001	I can analyze an absolute-value function using transformations.
M31-EXT-002	I can determine even or odd symmetry from a graph or formula.
M31-EXT-003	I can interpret a contextual logarithmic scale.

Math 32 Course at a Glance

Trigonometry Fundamentals

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Trigonometric Foundations

REQUIRED	
M32-FND-001	I can use the Pythagorean theorem to determine an unknown side of a right triangle.
M32-FND-002	I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
M32-FND-003	I can convert angle measures between degrees and radians.
M32-FND-004	I can determine coterminal and reference angles for a given angle.
M32-FND-005	I can calculate arc length from a central angle and radius.
M32-FND-006	I can determine exact coordinates on the unit circle.
M32-FND-007	I can evaluate all six trigonometric functions exactly for common angles.
M32-FND-008	I can solve right triangles using trigonometric ratios.
M32-FND-009	I can apply right-triangle trigonometry to contextual problems.
M32-FND-010	I can evaluate trigonometric functions from a point on a circle or terminal side.
M32-FND-011	I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED	
M32-GRF-001	I can graph basic sine and cosine functions using key features.
M32-GRF-002	I can identify amplitude, midline, period, and phase shift from an equation or graph.
M32-GRF-003	I can graph transformed sine and cosine functions.
M32-GRF-004	I can write sine and cosine equations that represent a periodic graph.
M32-GRF-005	I can construct a sinusoidal model from contextual information or data.
M32-GRF-006	I can interpret the parameters and outputs of a sinusoidal model in context.
M32-GRF-007	I can graph tangent, secant, and cosecant functions with their asymptotes.

Inverse Functions and Trigonometric Equations

REQUIRED	
M32-INV-001	I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
M32-INV-002	I can explain how restricted domains make inverse trigonometric functions possible.
M32-INV-003	I can solve trigonometric equations exactly or approximately on a specified interval.
M32-INV-004	I can express all solutions of a trigonometric equation in general form.
M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.

Math 32 Course at a Glance

Trigonometry Fundamentals

Non-Right Triangles

REQUIRED	
M32-TRI-002	I can solve a non-right triangle using the Law of Sines.
M32-TRI-003	I can determine the number of possible triangles in the ambiguous SSA case.
M32-TRI-004	I can solve a non-right triangle using the Law of Cosines.
M32-TRI-005	I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED	
M32-IDN-001	I can rewrite trigonometric expressions using reciprocal and quotient identities, including \`tan(theta) = sin(theta)/cos(theta)\`.
M32-IDN-002	I can rewrite trigonometric expressions using the Pythagorean identities.
M32-IDN-003	I can use even/odd and cofunction identities to rewrite trigonometric expressions.
M32-IDN-004	I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION	
M32-IDX-001	I can apply double-angle identities.
M32-IDX-002	I can apply half-angle identities.
M32-IDX-003	I can apply angle sum and difference identities.
M32-IDX-004	I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION	
M32-TRX-001	I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED	
M32-POL-001	I can plot and identify points in polar coordinates.
M32-POL-002	I can represent a polar point using equivalent coordinate pairs.
M32-POL-003	I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION	
M32-PLX-001	I can graph and interpret basic polar functions.
M32-PLX-002	I can convert equations between polar and rectangular forms.

Vectors

EXTENSION	
M32-VEC-001	I can represent a vector geometrically and in component form.
M32-VEC-002	I can add, subtract, and multiply vectors by scalars.
M32-VEC-003	I can determine vector magnitude, direction, and unit vector.
M32-VEC-004	I can resolve a vector into horizontal and vertical components.
M32-VEC-005	I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION	
M32-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M32-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M32-PAR-003	I can create or interpret parametric equations for motion in the plane.

MIDDLESEX SCHOOL | MATHEMATICS

Math 39 Course at a Glance

Precalculus with Data Analysis

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Linear Modeling and Residuals

REQUIRED	
M39-LIN-001	I can calculate and interpret average rate of change with appropriate units.
M39-LIN-002	I can construct a linear function from two data points, a table, a graph, or a context.
M39-LIN-003	I can fit a linear regression model to data using technology and interpret its parameters.
M39-LIN-004	I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
M39-LIN-005	I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
M39-LIN-006	I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED	
M39-EXP-001	I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
M39-EXP-002	I can construct an exponential function from two data points, a table, or a context.
M39-EXP-003	I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
M39-EXP-004	I can determine and interpret doubling time or half-life from an exponential model.
M39-EXP-005	I can solve exponential and logarithmic equations in modeling contexts.
M39-EXP-006	I can fit an exponential regression model using technology and evaluate its fit.
M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED	
M39-POW-001	I can identify a power function and interpret the parameters in y = kx^p.
M39-POW-002	I can construct a power function from two data points or a proportional relationship.
M39-POW-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M39-POW-004	I can recognize and model direct, inverse, and other power-variation relationships.
M39-POW-005	I can solve equations involving integer or rational powers.
M39-POW-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED	
M39-POL-001	I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
M39-POL-002	I can determine where a combination of functions is positive, zero, or undefined.
M39-POL-003	I can determine polynomial end behavior from degree and leading coefficient.
M39-POL-004	I can identify zeros and multiplicities and relate them to polynomial graph behavior.
M39-POL-005	I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
M39-POL-006	I can construct a polynomial function from specified zeros, multiplicities, and a point.
M39-POL-007	I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
M39-POL-008	I can build and optimize a polynomial model in a geometric or economic context.

Math 39 Course at a Glance

Precalculus with Data Analysis

Describing Data

REQUIRED

M39-STA-001	I can distinguish categorical and quantitative data.
M39-STA-002	I can construct and interpret frequency and relative-frequency tables.
M39-STA-003	I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
M39-STA-004	I can identify misleading features in a data display.
M39-STA-005	I can describe a distribution's modality, symmetry, skewness, and outliers.
M39-STA-006	I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
M39-STA-007	I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
M39-STA-008	I can construct and interpret a five-number summary and boxplot.
M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.
M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

M39-PRO-001	I can organize categorical data in a contingency table or Venn diagram.
M39-PRO-002	I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.
M39-PRO-003	I can identify an experiment, outcomes, events, and a sample space.
M39-PRO-004	I can calculate theoretical probabilities when outcomes are equally likely.
M39-PRO-005	I can use complements to calculate probabilities.
M39-PRO-006	I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.
M39-PRO-007	I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.
M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

M39-EV-001	I can calculate the expected value of a discrete probability situation.
M39-EV-002	I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

M39-POW-008	I can linearize power or exponential data and translate between the linearized and original models.

MIDDLESEX SCHOOL | MATHEMATICS

Math 49 Course at a Glance

Precalculus with Limits

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M49-FND-001	I can determine the domain and range of a function from a formula, graph, or table.
M49-FND-002	I can describe and apply transformations of a parent function.
M49-FND-004	I can analyze exponential functions using their equations, graphs, and contexts.
M49-FND-005	I can analyze logarithmic functions using their equations, graphs, and contexts.
M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED	
M49-FND-003	I can determine whether a function is even, odd, or neither.
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain.
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables.
M49-OPS-003	I can write a formula for the composition of two functions and determine its domain.
M49-OPS-004	I can decompose a function into a composition of simpler functions.
M49-OPS-005	I can determine whether a function has an inverse on a given domain.
M49-OPS-006	I can find and verify an inverse function algebraically.
M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables.

Power, Polynomial, and Rational Functions

REQUIRED	
M49-PWR-001	I can identify a power function and interpret the parameters in y = kx^p.
M49-PWR-002	I can construct a power function from two data points or a proportional relationship.
M49-PWR-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M49-PWR-004	I can recognize and model direct, inverse, and other power-variation relationships.
M49-PWR-005	I can solve equations involving integer or rational powers.
M49-PWR-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M49-PWR-008	I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
M49-ALG-002	I can predict the end behavior of a polynomial from its degree and leading coefficient.
M49-ALG-003	I can determine polynomial zeros and their multiplicities.
M49-ALG-004	I can construct a polynomial function from specified zeros or graphical features.
M49-ALG-005	I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
M49-ALG-006	I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
M49-ALG-007	I can sketch and analyze a rational function using its algebraic and asymptotic features.
M49-ALG-008	I can build or select a power, polynomial, or rational model for a context.

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Precalculus with Limits

Limits and Function Comparison

REQUIRED

M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input.
M49-LIM-002	I can estimate a limit from a graph or table.
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification.
M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior.
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior.
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

Sequences and Series

REQUIRED

M49-SEQ-001	I can evaluate and graph a sequence defined explicitly or recursively.
M49-SEQ-002	I can write explicit and recursive formulas for arithmetic sequences.
M49-SEQ-003	I can write explicit and recursive formulas for geometric sequences.
M49-SEQ-004	I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
M49-SEQ-005	I can create and interpret a sequence model for discrete change.
M49-SER-001	I can interpret and use summation notation.
M49-SER-002	I can calculate a finite arithmetic series.
M49-SER-003	I can calculate a finite geometric series.
M49-SER-004	I can determine whether an infinite geometric series converges and find its sum when it does.
M49-SER-005	I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists.
M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M49-PAR-003	I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form.
M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle.
M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse.
M49-CON-005	I can write, interpret, and graph implicit and parametric equations of a hyperbola.
M49-CON-006	I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
M49-CON-007	I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
M49-CON-008	I can explain a parabola as the set of points equidistant from a focus and directrix.